

Week 15: Package Management (Part 1)

Date: June 15, 2026

Study Time: 5-8 hours total

GOAL

Master apt package management on Ubuntu devices

STUDY TASKS (1-2 hours)

Reading

☐ **"How Linux Works"** - Chapter 8 (Software Installation - Debian)

Video Learning

☐ **LearnLinuxTV** - "APT Package Management"

☐ **Professor Messer** - "Linux+ Package Management"

Concepts

- ☐ What is apt vs apt-get?
 - ☐ Repositories and sources.list
 - ☐ Package dependencies
 - ☐ PPAs (Personal Package Archives)
 - ☐ dpkg (low-level package tool)
 - ☐ Package signing and GPG keys
 - ☐ Virtual packages
 - ☐ Package pinning
-

LAB WORK (3-4 hours)

Understanding apt

☐ **Check apt version**

```
bash
```

```
apt --version  
# apt is the newer, user-friendly command  
# apt-get is the older, script-friendly command
```

☐ View current repositories

```
bash
```

```
cat /etc/apt/sources.list  
  
# Also check additional sources:  
ls /etc/apt/sources.list.d/
```

Basic apt Commands

☐ Update package lists

```
bash
```

```
sudo apt update  
  
# This updates the LOCAL list of available packages  
# Does NOT install anything  
# Always run this before installing/upgrading
```

☐ Upgrade installed packages

```
bash
```

```
# Safe upgrade (doesn't remove packages)  
sudo apt upgrade  
  
# Full upgrade (may remove packages if needed)  
sudo apt full-upgrade  
  
# See what would be upgraded (dry run)  
apt list --upgradable
```

☐ Search for packages

bash

Search in package names and descriptions

```
apt search nginx
```

More specific search

```
apt search ^nginx$ # Exact match
```

List all available packages

```
apt list
```

List only installed packages

```
apt list --installed
```

☐ Show package information

bash

```
apt show nginx
```

Shows:

- Description

- Version

- Size

- Dependencies

- Homepage

- etc.

Installing Packages

☐ Install a package

bash

```
sudo apt install tree
```

Verify installation

```
tree --version
```

```
which tree
```

☐ Install multiple packages

bash

```
sudo apt install htop ncdu curl wget
```

☐ Install without prompting (for scripts)

bash

```
sudo apt install -y package_name  
# -y assumes "yes" to all prompts
```

☐ Install specific version

bash

```
# See available versions  
apt list -a package_name  
  
# Install specific version  
sudo apt install package_name=version
```

Removing Packages

☐ Remove a package

bash

```
sudo apt remove package_name  
# Removes package but keeps config files
```

☐ Completely remove (purge)

bash

```
sudo apt purge package_name  
# Removes package AND config files
```

☐ Remove unused dependencies

bash

```
sudo apt autoremove  
# Removes packages that were installed as dependencies  
# but are no longer needed
```

☐ Clean package cache

```
bash
```

```
# Remove downloaded package files
```

```
sudo apt clean
```

```
# Remove only outdated packages
```

```
sudo apt autoclean
```

Working with apt-cache

☐ Show package info (alternate method)

```
bash
```

```
apt-cache show nginx
```

```
# Similar to 'apt show' but different format
```

☐ Search package cache

```
bash
```

```
apt-cache search nginx
```

☐ Show dependencies

```
bash
```

```
apt-cache depends nginx
```

```
# Shows what nginx depends on
```

☐ Show reverse dependencies

```
bash
```

```
apt-cache rdepends nginx
```

```
# Shows what depends on nginx
```

☐ Package statistics

```
bash
```

```
apt-cache stats
```

```
# Shows total packages, versions, etc.
```

Working with dpkg

dpkg is the low-level package manager. apt uses dpkg under the hood.

☐ **List all installed packages**

```
bash

dpkg -l

# Search installed packages
dpkg -l | grep nginx
```

☐ **Show files installed by package**

```
bash

dpkg -L nginx
# Lists every file installed by nginx
```

☐ **Find which package owns a file**

```
bash

dpkg -S /usr/bin/nginx
# Shows: nginx: /usr/bin/nginx
```

☐ **Check if package is installed**

```
bash

dpkg -s nginx
# Shows package status and info
```

☐ **Install a .deb file directly**

```
bash

sudo dpkg -i package.deb

# If dependencies missing:
sudo apt install -f # Fix broken dependencies
```

Adding Repositories (PPAs)

 Only add trusted repositories!

☐ **Add a PPA**

bash

```
# Example: Add git PPA for latest version
sudo add-apt-repository ppa:git-core/ppa

# Update package list
sudo apt update

# Now install from PPA
sudo apt install git
```

☐ Remove a PPA

bash

```
sudo add-apt-repository --remove ppa:git-core/ppa
```

☐ List PPAs

bash

```
ls /etc/apt/sources.list.d/
```

Understanding Dependencies

☐ View dependency tree

bash

```
apt-cache depends --recurse --no-recommends nginx
```

☐ See why package is installed

bash

```
apt-cache rdepends package_name
```

☐ Simulate installation (see what would happen)

bash

```
apt install -s nginx
# -s = simulate
# Shows what would be installed without actually doing it
```

Practical Exercises

☐ Install useful tools on your Ubuntu laptop

```
bash

sudo apt update
sudo apt install -y \
    htop \           # Better top
    ncddu \          # Disk usage analyzer
    tree \           # Directory tree viewer
    curl \           # Transfer data
    wget \           # Download files
    net-tools \       # ifconfig, netstat
    vim \            # Text editor
    git \            # Version control
    tmux             # Terminal multiplexer
```

☐ Check what's installed

```
bash

dpkg -l | grep htop
which htop
htop --version
```

☐ Practice removing

```
bash

# Install a test package
sudo apt install cowsay
cowsay "Hello!"

# Remove it
sudo apt remove cowsay

# Verify removal
which cowsay # Should return nothing
```

Package Management Script

☐ Create package management script

bash

```
nano ~/apt-report.sh
```

```
#!/bin/bash
```

```
echo "=== APT Package Report ==="
```

```
echo "Generated: $(date)"
```

```
echo ""
```

```
echo "--- Installed Packages ---"
```

```
dpkg -l | grep ^ii | wc -l
```

```
echo ""
```

```
echo "--- Upgradable Packages ---"
```

```
apt list --upgradable 2>/dev/null | grep -v "Listing" | wc -l
```

```
echo ""
```

```
echo "--- Held Packages ---"
```

```
dpkg -l | grep ^h | wc -l
```

```
echo ""
```

```
echo "--- Broken Packages ---"
```

```
dpkg -l | grep ^i[^i] | wc -l
```

```
echo ""
```

```
echo "--- Disk Usage by Package Cache ---"
```

```
du -sh /var/cache/apt/archives/
```

```
echo ""
```

```
echo "--- Recently Installed (last 10) ---"
```

```
grep " install " /var/log/dpkg.log | tail -10
```

```
chmod +x ~/apt-report.sh
```

```
./apt-report.sh
```

Troubleshooting

☐ Fix broken packages

bash

```
sudo apt install -f
```

```
# or
```

```
sudo apt --fix-broken install
```

☐ Reconfigure package

bash

```
sudo dpkg-reconfigure package_name
```

☐ Clear package locks (if stuck)

bash

```
# Only if you're sure no apt is running!
sudo rm /var/lib/dpkg/lock-frontent
sudo rm /var/lib/dpkg/lock
sudo dpkg --configure -a
```

FLASHCARD COMMANDS

1. `apt update`
2. `apt upgrade`
3. `apt install`
4. `apt remove`
5. `apt purge`
6. `apt search`
7. `apt show`
8. `apt autoremove`
9. `apt clean`
10. `apt-cache`
11. `dpkg -l`
12. `dpkg -L`
13. `dpkg -S`
14. `add-apt-repository`

Concepts:

- Difference between apt and apt-get?
 - What does apt update do?
 - What does apt upgrade do?
 - What's a PPA?
 - What's a dependency?
 - What's dpkg?
 - Where are packages stored?
-

EXAM OBJECTIVES

[https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant: 2.3 Given a scenario, create, modify, and redirect files

NOTES

Packages installed this week:

PPAs added:

Common commands used:

Issues encountered:

Completed: ☐

QUICK REFERENCE

Common apt Workflow:

```
bash

sudo apt update           # Update package lists
sudo apt upgrade          # Upgrade packages
sudo apt install package  # Install package
sudo apt remove package   # Remove package
sudo apt autoremove       # Clean up
sudo apt clean            # Clear cache
```

apt vs apt-get:

Command	apt	apt-get
Update lists	apt update	apt-get update
Upgrade	apt upgrade	apt-get upgrade
Install	apt install	apt-get install
Remove	apt remove	apt-get remove
Search	apt search	apt-cache search
Show info	apt show	apt-cache show
User-friendly?	✓ Yes	No
Use in scripts?	Either	✓ Preferred

Important Directories:

- `/etc/apt/sources.list` - Main repository list
- `/etc/apt/sources.list.d/` - Additional repositories
- `/var/cache/apt/archives/` - Downloaded .deb files
- `/var/lib/apt/lists/` - Package lists
- `/var/lib/dpkg/` - dpkg database

Package States:

- ii = Installed
- rc = Removed (config files remain)
- un = Not installed
- hi = Held (prevented from upgrading)

Previous Week: Bash Scripting & Cron

Next Week: Package Management (dnf/rpm)